

Key: Older Than They Used to Be

84-355 Democracy's Data | Officeholder age | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Structure. The brief (`officeholder-age-brief.Rmd`) is now a chapter, “Older Than They Used to Be”: it is organised by argument under prose headings and carries five numbered figures rather than numbered steps — Figure 1, two and a third centuries of congressional age; Figure 2, the same 68 years against two rows of one government table; Figure 3, the mean age split into its two components; Figure 4, the full age distribution by era; Figure 5, age on arrival against length of career. The lab (`officeholder-age.Rmd`) is unchanged and still runs Steps 1–9 with the code exposed. **Students should read the brief first. Step numbers below refer to the lab.** The two cover the same ground in the same order, so each lab Step has a chapter heading facing it: Step 1 = “One column, and what a birth year buys you”; Step 2 = “When it happened” (Figure 1); Step 3 = “Everyone lives longer now”; Step 4 = “The row further down the same page” (Figure 2); Step 5 = “Arriving older, or staying longer” (Figure 3); Step 6 = “The median hides the tail” (Figure 4); Step 7 = “The careers you cannot see” (Figure 5); Step 8 = “Who leaves the data before you can count them”; Step 9 = “What this evidence will bear”.

This is the course’s cleanest example of a denominator flipping a conclusion. Everything else in the session is context for Step 4. If the room leaves with one thing, it should be that a comparison is only as good as what you compared to, and that the wrong choice here is the one everybody makes.

Every number below was recomputed from the built files when this was knitted.

The session in one sentence

Congress is at or near its oldest ever, the popular rebuttal — *people live longer now* — collapses the moment you use remaining life expectancy at 65 instead of life expectancy at birth, and most of the increase turns out to come from members arriving older rather than from incumbents staying.

The findings

```
data.frame(  
  quantity = c("median age, Congress, now", "median age, Senate, now",  
               "Congress median age gain, 1950-2018",  
               "life expectancy at birth, gain", "remaining life at 65, gain",
```

```

      "% of the headline gain that is life after 65",
      "% of the House rise from arriving older",
      "% of the Senate rise from arriving older",
      "% of Congress over 70 now"),
value = c(val("Congress", NOW), val("Senate", NOW), A1 - A0,
      E0b - E0a, E6b - E6a,
      round(100 * (E6b - E6a) / (E0b - E0a)),
      round(100 * unname(DH["entry"] / DH["total"])),
      round(100 * unname(DS["entry"] / DS["total"])),
      round(val("Congress", NOW, "pct_70plus"), 1)),
row.names = NULL)

```

```

#>                                quantity value
#> 1                median age, Congress, now  59.0
#> 2                median age, Senate, now   65.0
#> 3      Congress median age gain, 1950-2018    7.0
#> 4      life expectancy at birth, gain   10.5
#> 5      remaining life at 65, gain     5.6
#> 6 % of the headline gain that is life after 65  53.0
#> 7      % of the House rise from arriving older  68.0
#> 8      % of the Senate rise from arriving older  43.0
#> 9                % of Congress over 70 now   24.1

```

- **Median member 59, median senator 65.** The three oldest Congresses on record are 2017, 2019, 2021.
- **The rise is recent.** The median member was 50 in 1979, a figure Congress had not touched since 1913.
- **Congress aged 7 years, 1950 to 2018. Life expectancy at birth rose 10.5; remaining life expectancy at 65 rose 5.6.** The first comparison says Congress aged slower than the country, the second says faster.
- **At most 53% of the famous life expectancy gain is about people who have already reached 65.**
- **63% of Congress's rise in mean age is arrival age,** not tenure — and it splits 68% House against 43% Senate.
- **24.1% of Congress is over 70,** and 35.9% of the Senate.

The three things to get said

1. Step 4 is the session. Two numbers from the same NCHS table, one section apart, give opposite answers to the same question. **Take a vote before you show the second one.** Ask the room: life expectancy rose 10.5 years since 1950 and Congress aged 7 — has Congress aged more or less than the country? Almost everyone says less, correctly, using the wrong number. Then show 5.6.

The reason it works is that **nobody is being dishonest in the wrong version.** Life expectancy at birth is the number everyone means by “life expectancy”. It is published, official, correct, and irrelevant, and the only way to know that is to ask what population it describes.

2. The decomposition is the surprise. Ask before revealing: *is Congress older because incumbents stay longer, or because new members arrive older?* The room will say incumbency, essentially unanimously. It is 63% arrival age. **And press on the chamber split** — the House is 68% arrival age and the Senate only

43%. One mechanism, two institutions, two different answers, which is a good argument against explaining “Congress” as a single thing.

3. The rise is recent, and it reverses. Congress got *younger* through the 1960s and 1970s (Step 5’s era table: mean age -2.7 years over 1950-1980) and then rose sharply. A story about slow institutional decay does not fit. Whatever changed, changed around 1980, and this data does not say what.

The line for the board: *the comparison is where the argument is hiding.*

Walking the steps

Step	What it does	Min	Watch for
1 One file, one column	born is a year; +/- 1 on every age	6	Show the Biden row. 78 in life, 79 by arithmetic. Then Cleveland, half-missing. Students read the long-run chart as a steady climb. It is not; make them find the trough. Let the wrong answer win here. Do not signpost. The pivot. Vote first. Then the sparse-series and two-sources caveats. Predict before revealing. The chamber split is the follow-up. Why is the median the wrong summary here? The pipeline result runs the wrong way; students assume it explains the Senate. This one cuts <i>against</i> the lab’s own headline. Say so. “Why” and “does it matter” are both outside this data.
2 How old is Congress	Median 50 in 1979, 60 at peak	7	
3 The rebuttal	Life expectancy up 10.5, Congress up 7	5	
4 The right denominator	e(65) up 5.6	14	
5 Entering vs staying	63% arrival age	10	
6 The tail	24.1% over 70	6	
7 Careers and censoring	Right-censoring, and the House-to-Senate pipeline	8	
8 Deaths in office	Rate fell from 3.2% to 0.6%	5	
9 What you can say	Including what you cannot	5	

If you are short on time, protect Steps 3, 4 and 5. Steps 3 and 4 are one move and must not be split across a break. Step 5 is what stops the session being a single trick.

The thing that makes Step 4 work, and how not to spoil it

Step 3 is written to be persuasive. Read it aloud in the voice of somebody who thinks they have won the argument, because that is the voice students will meet it in outside the classroom.

The mechanism, if somebody asks for it in more detail than the lab gives: life expectancy at birth is an average over a whole synthetic cohort, and in 1900 that average was dragged down by an enormous number of deaths before age 5. Removing those deaths raises the average by a lot without changing anybody's prospects at 60. **The quantity that changed least across the twentieth century is exactly the quantity that describes a senator.**

Anticipate the good objection, because a sharp student will make it: $e(65)$ is conditional on reaching 65, and members of Congress are not a random draw from the population — they are richer, better educated and better insured, so their own remaining life expectancy is higher than the national $e(65)$ and probably rose faster. That is correct. It makes the lab's comparison *conservative* in one direction and unmeasurable in another, and the honest answer is that the data here cannot settle it. **A student who raises this unprompted has understood the step better than the step does.**

What to do about the numbers that cut the other way

Two results in this lab work against its own headline, and leaving them in is deliberate. Do not skip them.

- **Step 8, deaths in office.** Members used to die in office much more often: 3.2% of member-terms between 1900 and 1949 ended in a death, against 0.6% since 2000. Every such death removes an old member from every later Congress, so part of the apparent youth of historical Congresses is survival, not preference. This is the one place where “people live longer now” genuinely does explain something.
- **Step 7, the pipeline.** Senators who came up through the House arrive *younger* (51.6) than those who did not (53.3). The obvious explanation for an older Senate is wrong.

Model the behaviour you want. A lab that only reports the results supporting its thesis teaches the wrong thing regardless of how good the thesis is.

Option A — Use the wrong denominator on purpose

Verified output

```
E7a <- le$e75[le$year == 1980]; E7b <- le$e75[le$year == EY]
data.frame(
  yardstick = c("life expectancy at birth", "remaining life at 65",
               "remaining life at 75 (1980-2018 only)"),
  gain = c(E0b - E0a, E6b - E6a, E7b - E7a),
  congress_gain = c(A1 - A0, A1 - A0,
                   A1 - val("Congress", 1980)),
  congress_faster = c((A1 - A0) > (E0b - E0a), (A1 - A0) > (E6b - E6a),
                    (A1 - val("Congress", 1980)) > (E7b - E7a)))
```

```
#>                                yardstick gain congress_gain congress_faster
#> 1                life expectancy at birth 10.5                7          FALSE
#> 2                remaining life at 65   5.6                7          TRUE
#> 3 remaining life at 75 (1980-2018 only)  1.9                10          TRUE
```

3/4: Runs all three, reports that the answer changes, picks one.

4/4: Argues for e(65) on the grounds that it matches the age of the population being described, and notices that e(75) is even more conservative but covers a shorter window, so the two are not interchangeable.

4/4, best: Says that no single yardstick is right, because the question “has Congress aged relative to longevity” is really several questions, and picks the one that matches the claim being tested — then writes a sentence that carries both numbers so the reader can check. **The sophisticated answer is that the choice should follow the claim, not the other way round.**

Option B — Pick a different window

Verified output

```
w <- function(y1, y2) {
  a <- val("Congress", y1); b <- val("Congress", y2)
  c(congress = b - a,
    e0 = le$e0_hus[le$year == y2] - le$e0_hus[le$year == y1],
    e65 = le$e65[le$year == y2] - le$e65[le$year == y1])
}
rbind(`1950-1980` = w(1950, 1980), `1980-2018` = w(1980, EY),
      `1950-2018` = w(1950, EY))
```

```
#>      congress    e0 e65
#> 1950-1980      -3  5.5 2.5
#> 1980-2018      10  5.0 3.1
#> 1950-2018       7 10.5 5.6
```

This one has a sharp answer and you should know it before you assign it. The two halves do not merely differ, they point opposite ways: over 1950-1980 Congress’s median age **fell** while both life-expectancy series rose, and over 1980-2018 Congress’s median age rose 10 years against 5.0 at birth and 3.1 at 65.

3/4: Reports the two halves and notices they differ.

4/4: Sees that the entire story is post-1980, and that in the first half Congress aged more slowly than longevity on *either* yardstick.

4/4, best: Notices the sting — **after 1980 Congress outpaces life expectancy at birth as well**, so on the modern window you do not need the careful denominator to win the argument, and the careful denominator is what lets you make the claim about the *whole* period. Then observes that the choice of window is itself a denominator-like choice which the lab made for a data-availability reason rather than a substantive one. That is the best available answer and it is a criticism of the lab.

Option C — One state, or one party

Verified output

```
hs <- m[m$chamber %in% c("House", "Senate") & !is.na(m$age) &
      m$party_code %in% c(100, 200), ]
hs$party <- ifelse(hs$party_code == 100, "Democrat", "Republican")
py <- tapply(hs$age, list(hs$congress, hs$party), median)
py <- py[rownames(py) %in% as.character(c(sitting(1950), sitting(1980),
                                          sitting(2000), sitting(EY),
                                          sitting(NOW))), ]

round(py, 1)
```

```
#>      Democrat Republican
#> 81      51.0         55
#> 96      50.5         50
#> 106     54.0         54
#> 115     63.0         57
#> 119     59.0         59
```

3/4: Computes a party or state series and reports a gap.

4/4: Reports the gap *and* its size relative to year-to-year wobble, and notices that a state delegation of two senators and a handful of representatives cannot support a median at all.

4/4, best: Recognises that party age differences move with which party is winning open seats — a party that just had a good year is younger because it has more freshmen — so **the party gap is partly a measure of recent electoral success, not of anything about the parties.** Anyone who says that has transferred the composition lesson from **regional-shift** without being told to.

Option D — The decomposition, one chamber at a time

No fully verified output; grade the reasoning.

3/4: Runs the decomposition separately and reports that the chambers differ.

4/4: Locates roughly when they diverged and offers a mechanism tied to institutional differences — six-year terms, statewide constituencies, the cost of a Senate race — rather than to a single event.

4/4, best: Notices that “entry age” for the Senate is measured from Senate entry, so a House veteran’s earlier career is invisible to it, and asks what the decomposition would look like measured from first federal office. **That is a real limitation of the built data and the student found it in the code.**

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is given.
- **The move that earns a 4** is treating the choice of comparison as the substantive act it is, rather than as a preliminary to the analysis.
- **A 2** is “Congress is old and that’s bad.” The lab is explicitly not an argument for term limits or an age cap, and an essay that turns it into one has skipped the entire method.
- **Also a 2:** reporting the 7-year rise without the denominator discussion at all.
- **Reward anyone who notices what is missing:** candidates who lost, any measure of performance, and remaining life expectancy for people like members of Congress rather than for the country. All three are named in the lab and the good answers go further than the naming.
- **Reward anyone who reports a number that cuts against them** — Step 8 is sitting there for exactly this.

Anticipated questions

- *“Isn’t age 65 arbitrary?”* — Yes. It is what NCHS publishes, and 75 is the other option, and the lab shows both. The defensible principle is to condition on an age the population in question has reached; 65 is close to the median member and 75 is not, which is a reason to prefer it here.
- *“Why not compare to the median age of the country?”* — Better question than it sounds, and it is a different comparison: the country’s median age moves with the birth rate, so it would be measuring demography rather than longevity. The lab does not have that series, and saying so is the honest answer.
- *“Shouldn’t we just use members’ own life expectancy?”* — Yes, and nobody publishes it. See the objection in the Step 4 section above.
- *“Isn’t this an argument for an age limit?”* — Not from this data. The lab establishes what is true about officeholder age and explicitly cannot speak to performance. **Say that plainly once and do not relitigate it;** the question will come up in every section.
- *“Why is born only a year?”* — Because the file was built for roll-call analysis, where a birth date is not needed. **A dataset’s precision follows its original purpose, and you are almost always using it for something else.**

Connections

- **dw-nominate** — the same Voteview file, a completely different derived quantity. Worth saying out loud: **one carefully built file answers many unrelated questions**, and most of the work in this lab was choosing what to compute, not finding data.
- **scdb** — the Supreme Court, where life tenure makes this question sharper and the sample size makes it harder. Run after this one.
- **house-competition** — uncontested races and incumbency advantage, which is the plausible mechanism behind the “staying longer” half of Step 5. Students who did that lab will reach for it; the decomposition tells them how much of the effect it could possibly explain.
- **apportionment** — the other lab where a technical choice nobody argues about turns out to determine something people care about.